

Analysis of FlexCast Reconfiguration

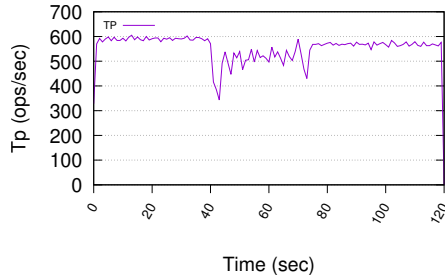
1 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 2 min
- Reconfig interval: 30sec
- Regions/Nodes: 3
(Calif., Virg., Sao Paulo)
- Clients: 150 (50 per region)
- Locality: 95%

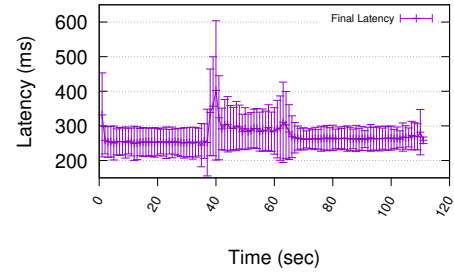


2 Results

2.1 Throughput



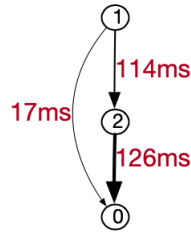
2.2 Latency



2.3 Reconfiguration details

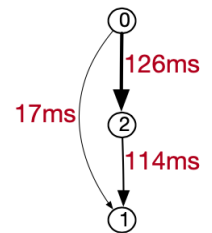
```
workload 1 : [
  {
    "destination": [0, 1],
    "percentage": 0.032794576
  },
  {
    "destination": [0, 2],
    "percentage": 0.57712483
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.050535202
  },
  {
    "destination": [1, 2],
    "percentage": 0.3395454
  },
]
```

Took 55 ms

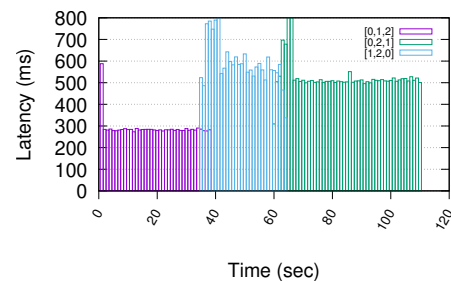
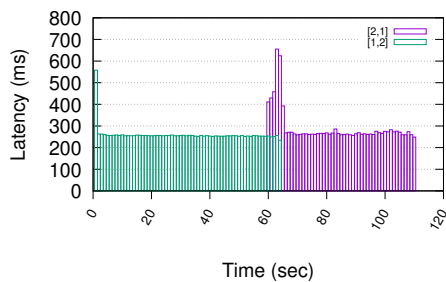
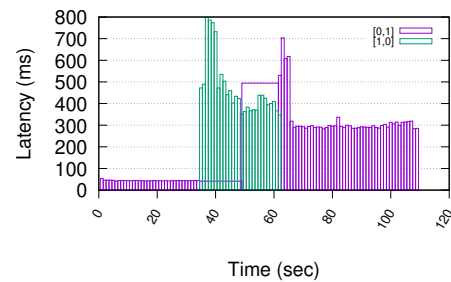
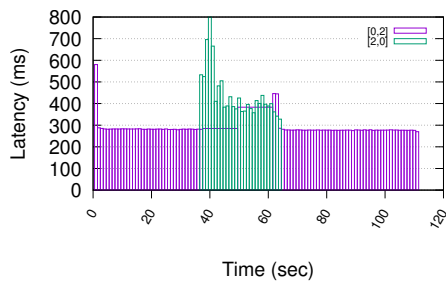


```
workload 2 : [
  {
    "destination": [1, 2, 0],
    "percentage": 0.05174473
  },
  {
    "destination": [2, 0],
    "percentage": 0.5442484
  },
  {
    "destination": [1, 0],
    "percentage": 0.034562822
  },
  {
    "destination": [1, 2],
    "percentage": 0.36944407
  },
]
```

Took 5 ms

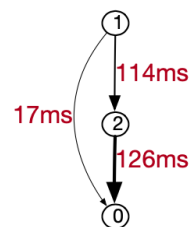


2.4 Latencies per dests (90p)



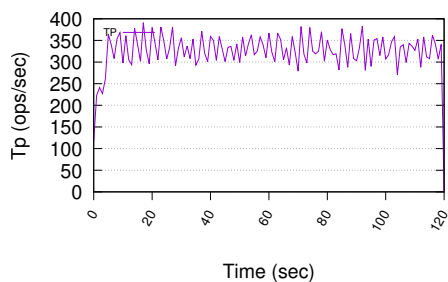
3 Experiment Config.

- TPCC/AWS Latencies
- Regions/Nodes: 3
(Calif., Virg., Sao Paulo)
- Exp. Duration: 2 min
- Clients: 150 (50 per region)
- Reconfig interval: no reconfig
- Locality: 95%

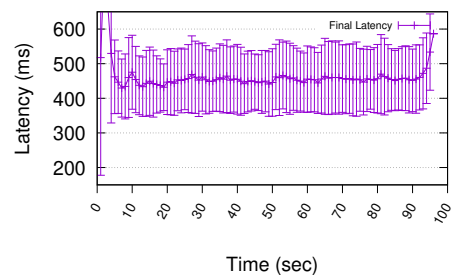


4 Results

4.1 Throughput

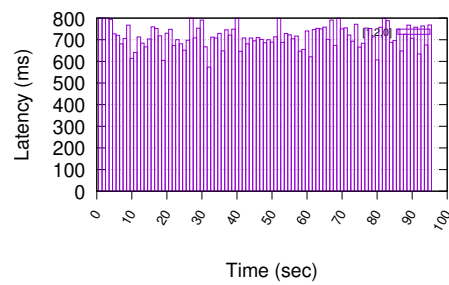
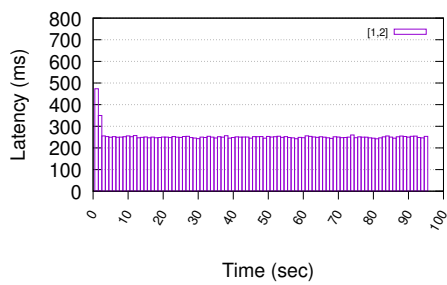
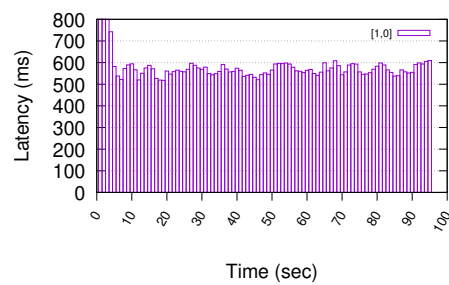
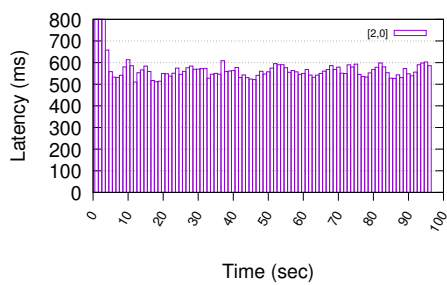


4.2 Latency



4.3 Reconfiguration details

4.4 Latencies per dests (90p)



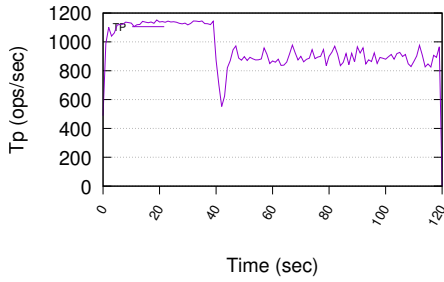
5 Experiment Config.

- **NO CLIENT LATENCIES**
- **TPCC/AWS Latencies**
- **Exp. Duration:** 2 min
- **Reconfig interval:** 30sec
- **Regions/Nodes: 3**
(Calif., Virg., Sao Paulo)
- **Clients: 150** (50 per region)
- **Locality: 95%**

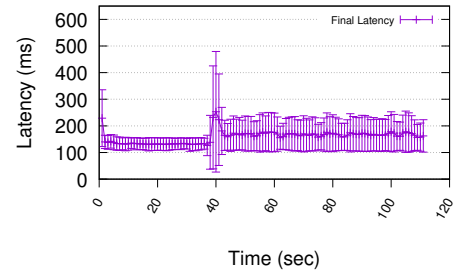


6 Results

6.1 Throughput



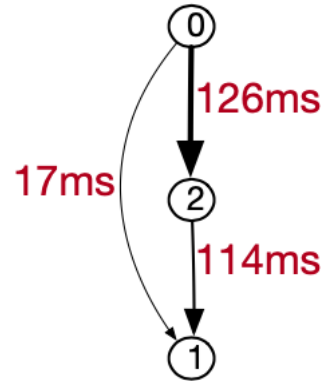
6.2 Latency



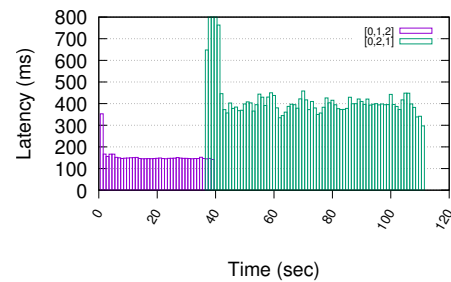
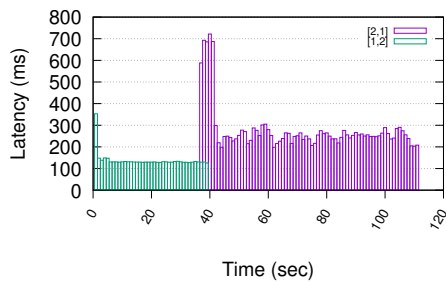
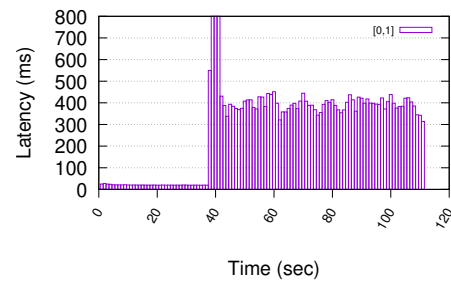
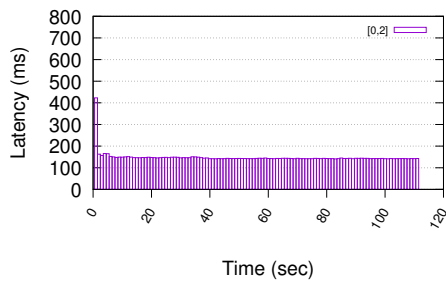
6.3 Reconfiguration details

```

workload 1 : [
  {
    "destination": [0, 1],
    "percentage": 0.03268678
  },
  {
    "destination": [0, 2],
    "percentage": 0.5705819
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.052015986
  },
  {
    "destination": [1, 2],
    "percentage": 0.34471536
  }
]
Took 55 ms
  
```

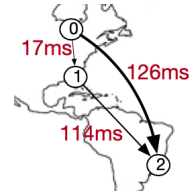


6.4 Latencies per dests (90p)



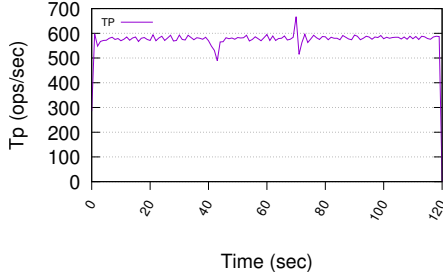
7 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 2 min
- Reconfig interval: 30sec
- Regions/Nodes: 3
(Calif., Virg., Sao Paulo)
- Clients: 150 (50 per region)
- Locality: 99%

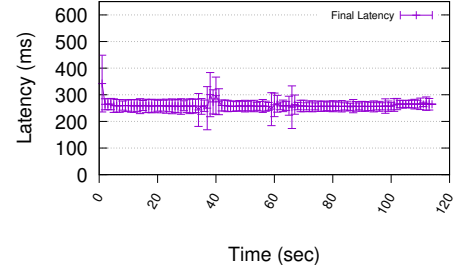


8 Results

8.1 Throughput



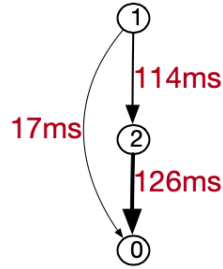
8.2 Latency



8.3 Reconfiguration details

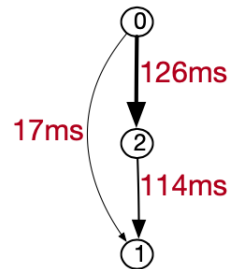
```

workload 1 : [
  {
    "destination": [0, 1],
    "percentage": 0.007280495
  },
  {
    "destination": [0, 2],
    "percentage": 0.6292179
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.01229401
  },
  {
    "destination": [1, 2],
    "percentage": 0.3512076
  },
]
Took 55 ms
  
```

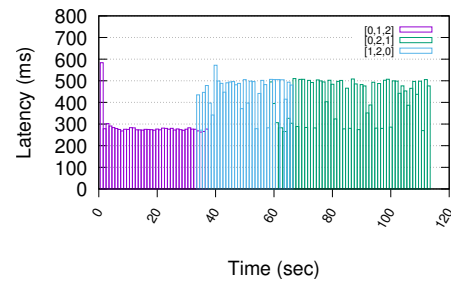
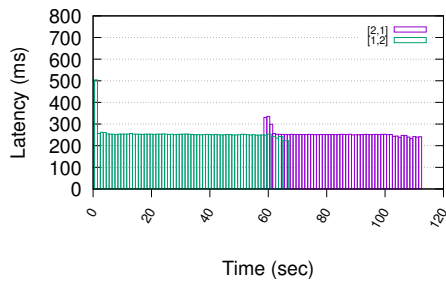
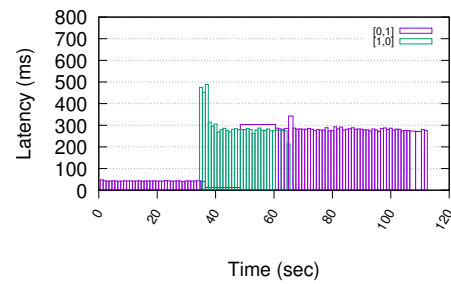
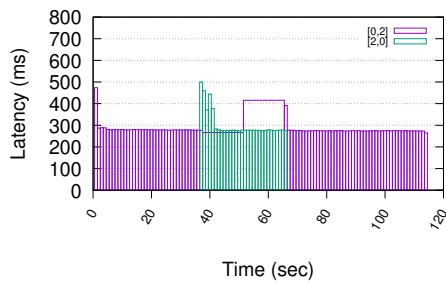


```

workload 2 : [
  {
    "destination": [1, 2, 0],
    "percentage": 0.010980073
  },
  {
    "destination": [2, 0],
    "percentage": 0.6253994
  },
  {
    "destination": [1, 0],
    "percentage": 0.006855284
  },
  {
    "destination": [1, 2],
    "percentage": 0.35676524
  },
]
Took 55 ms
  
```

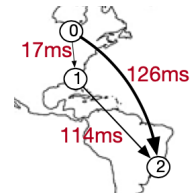


8.4 Latencies per dests (90p)



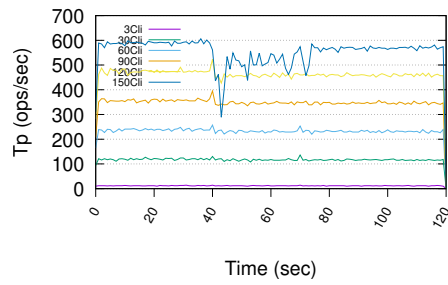
9 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 2 min
- Reconfig interval: 30sec
- Regions/Nodes: 3 (Calif., Virg., Sao Paulo)
- Clients: 3 to 150 (1/3 per region)
- Locality: 95%

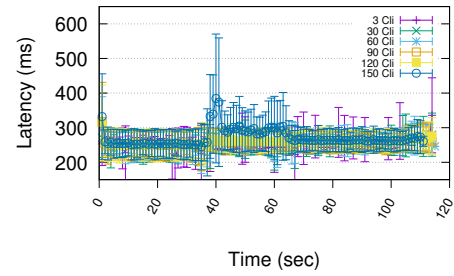


10 Results

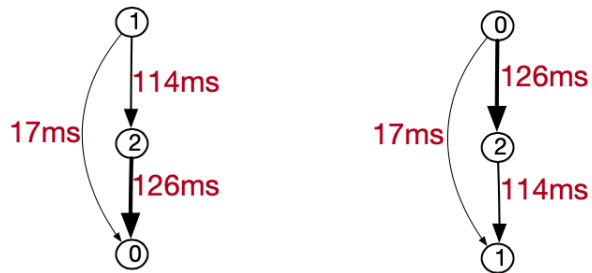
10.1 Throughput



10.2 Latency

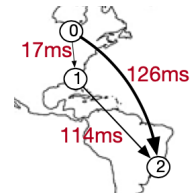


10.3 Reconfiguration details



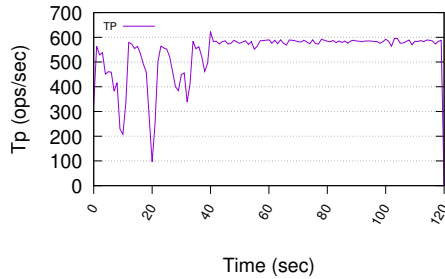
11 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 2 min
- Reconfig interval: 30sec
- Regions/Nodes: 3
(Calif., Virg., Sao Paulo)
- Clients: 150 (1/3 per region)
- Locality: 100%
- GC: 10sec

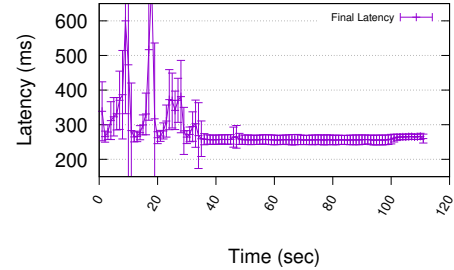


12 Results

12.1 Throughput



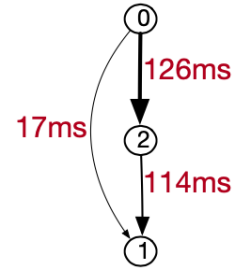
12.2 Latency



12.3 Reconfiguration details

```
"workload": [
  {
    "destination": [0, 2],
    "percentage": 0.64473253
  },
  {
    "destination": [0, 1, 2],
    "percentage": 2.2062879E-4
  },
  {
    "destination": [1, 2],
    "percentage": 0.35504687
  }
]
```

Took 35 ms



12.4 Latencies per dests (90p)

